

Semester 2

Course Code	19U2CRCMT2
Title of the course	Analytic Geometry, Theory of Equations and Numerical methods
Semester in which the course is to be taught	2
No. of credits	3
No. of contact hours per week	4
Total Hours	72

Course Objectives

- ☐ to explain more ideas of conics;
- ☐ Introduce theory of equations
- ☐ Introduce numerical methods

Course Outcomes

On completion of this course, successful students will be able to:

- ☐ find the equation to tangent, normal at a point on a conic ;
- ☐ find the polar equation of a line, circle , tangent and normal to conics

Text Books:

1. Manicavachagom Pillay , Natarajan – Analytic Geometry (Part I, Two Dimensions)
2. Engineering Mathematics, N.P. Bali, Manish Goyal, Lakshmi publications

MODULE I

(25hrs)

Tangents and Normals (parametric form only) of a conic, Orthoptic locus. Pole and Polar. Chord in terms of given points. Conjugate diameters of ellipse and hyperbola. Asymptotes of a hyperbola, conjugate hyperbola and rectangular hyperbola. (Relevant sections of Text 1)

MODULE II

(10hrs)

Polar co-ordinates, polar equation of a line, polar equation of a circle and polar equation of a conic. Polar equations of tangent and normal to these curves. (Relevant sections of Text 1)

MODULE III

(18 hrs)

Zero of a Polynomial, Division Algorithm, Polynomial Equation, Root of an Equation, Synthetic Division, Fundamental Theorem of Algebra, Multiplication of Roots, Diminishing and increasing the roots, Removal of Terms, Reciprocal Equations, Sum of the Integral Powers of the Roots and Symmetric Functions, Symmetric Functions of the Roots, Descarte's Rule of Signs, Cardan's Method, Descarte's Method, Ferrari's Solution of the Biquadratic. (Sections 2.5-2.18 and 2.20-2.21 of Text 2)

Module IV

(19 hrs)

Finite Difference and Interpolation

Forward and backward operators, Shift operator, its properties, Newton Gregory Formula, Lagrange's formula, Gauss formula, Stirling formula, Laplace Formula, Newton divided difference formula, Numerical integration and differentiation. Newton cotes, Trapezoidal and Simpson's formula (sections 22.1 to 22.11(d))

References

- 1.S.K . Stein – Calculus and analytic Geometry , (McGraw Hill)
- 2.A. N. Das – Analytic Geometry of two and three dimension (New Central Books)
- 3.Thomas and Finney - Calculus and analytical geometry (Addison-Wesley)
4. Quazi Shueb Ahamad - Numerical and Statistical Techniques (Ane Books)